Biospectra : Vol. 11(1), March, 2016, pp. 45-48

An International Biannual Refereed Journal of Life Sciences

Plant Science

ISSN : 0973-7057

Drought stress leads to alteration of protein spots in rice, *Oryza sativa* L.

Aparna Jha[a] & Yashodhara Verma[b]

[a]Jacob School of Biotechnology & Bioengineering, SHIATS,
(Formerly AAIDU), Allahabad, U.P

Received : 11th January, 2015; Revised : 9th February, 2016

Abstract : Apart from the regular circadian and seasonal perturbations, there may be certain other rapid and unpredicted disturbances in the environment resulting in stressful conditions. For instance, paucity of water for long periods due to lack of irrigation, infrequent rains or lowering of water table results in drought stress. Plants can initiate a number of molecular, cellular, and physiological changes to respond and adapt to drought stress, thus enabling them to survive. Synthesis of stress proteins is a ubiquitous response to cope with prevailing stressful conditions. We constructed here a proteome map to investigate the effect of drought stress on the rice (*Oryza sativa* L.) proteome. Plants were stressed for 24, 48, 72 and 96 hours. Total proteins were extracted and separated by two dimensional gel electrophoresis followed by Coomassie staining.

Keywords : Drought stress, protein spots altration, *Oryza sativa* L.

INTRODUCTION

Abiotic stresses such as drought are very important factors that endanger the crop production and stability (**Zombori et al, (2008)**). Rice is a notoriously drought-susceptible crop due in part to its small root system, rapid stomatal closure and little circular wax during mild water stress (**Hirasawa, 1999**). Reduction of photosynthetic activity, accumulation of organic acids and osmolytes, and changes in carbohydrate metabolism, are typical physiological and biochemical responses to drought stress (**Tabaeizadeh, 1998**). Drought affects both elongation and expansion growth **(Anjum *et al.*, 2003a; Bhatt & Srinivasa Rao, 2005; Kusaka *et al.*, 2005; Shao *et al.*, 2008).** Drought greatly suppresses cell expansion and cell growth due to the low turgor pressure.

The responses of plants to various abiotic stresses have been important subjects of physiological **(Levitt, 1980),** molecular and transgenic studies **(Bajaj et al., 1999; Hasegawa et al., 2000; Zhang et al., 2000)**. The identification of novel genes, determination of their expression patterns in response to the stresses, and an improved understanding of their functions in stress adaptation will provide us the basis of effective engineering strategies to improve stress tolerance **(Cushman and Bohnert, 2000).** A number of genes have been reported to be induced by drought and their products are thought to function in stress tolerance and response **(Bray, 1997; Thomashow, 1999; Shinozaki and Yamaguchi-Shinozaki, 2000).** Synthesis of stress proteins is a ubiquitous response to cope with prevailing stressful conditions. Most of the stress proteins are soluble in water and therefore contribute towards the stress tolerance phenomena by hydration of cellular structures **(Wahid *et al.*, 2007).** Both quantitative and qualitative changes of proteins were detected during water stress (**Riccardi *et al.*, 1998**). Evidence is increasing in favor of a relationship between the accumulation of drought-induced proteins and physiological adaptations to water limitation (**Bray, 1993; Han** and **Kermode, 1996; Riccardi *et al.*, 1998**).

*Corresponding author :
Phone : 09431772796
E-mail : namit12312@gmail.com

MATERIALS AND METHODS

The present study was implicated with identifying

the proteins that are specifically altered in rice (Oryza sativa L.) during drought stress and during recovery from the stress. Seeds of Sahabhagid dhan were washed thoroughly in tap water followed by detergent for 10 min. The seeds were further treated with (30% v/v) Sodium hypochlorite solution for 30 minutes. The surface sterilized seeds were shown on absorbent cotton pads placed in plastic trays and allowed to germinate. Four day old rice seedlings were subjected to drought stress and transferred to control condition for recovery. Stress was given for different duration of time (24, 48, 72 and 96 hours) and after that, the seedlings were recovered for the same period. Samples were harvested after each stress treatment and recovery.

Samples for 2D analysis were prepared following the procedure of **Dubey *et al.*, 2003.** Shoots of rice seedlings were homogenized in liquid Nitrogen up to the formation of fine powder. Polyvinyl poly pyrrolidone (PVPP) was added to the samples (50 mg/ g fresh weight of the tissue) during homogenization. The powder was then dissolved in **(Zivy *et al.*, 1983)** buffer containing 30 mM Tris-Hcl (pH 8.8), 1 mM L-ascorbic acid, 1 mM EDTA-Na (Ethylenediaminetetraacetic acid), 5 mM $MgCl_2$, 1 mM dithiothreitol (DTT) and 1 mM phenyl methyl sulphonyl fluoride (PMSF). This extract was centrifuged at 10,000 rpm for 15 min at 4°C and the supernatant was transferred to eppendorf tubes. Then the soluble protein extract was precipitated in TCA For 1 hr. It was again centrifuged at 14,000 rpm for 20 min at 4°C and the pellet, which contained the total protein, was washed with Acetone for 30 minutes. This solution was again centrifuged at 14,000 rpm for 10 min at 4°C and the pellet was dissolved in 150mL Reswelling buffer **(Pandey *et al.*, 2006).** Iso-electric focusing (IEF) of approximately 500Mg of protein reswelling buffer (8M urea, 2! w/v CHAPS, 50 Mm DTT, 0.2! Bio-Lyte® 3/10Ampholytes and 0.01! w/v bromophenol blue) was run using immobilized pH gradient strips for 12 h at 50 V, 15minat 250 V, 2h gradient to4000 V, and hold at500 V until the run is stopped. After IEF, IEF strips were equilibrated for 10 min in SDS-PAGE equilibration buffer É (6 M urea, 0.375 M Tris, pH 8.8, 2! SDS, 20! glycerol, 2! w/v DTT) followed by SDS-PAGE equilibration buffer ÉÉ (6 M urea, 0.375 M Tris, pH 8.8, 2! SDS, 20! glycerol, 2.5! w/v iodoacetamide). Second dimension SDS-PAGE was run in BioRad Protein® ÉÉ ×i Cell according to manufacturer recommendations. Gels were transferred in fixation solution (methanol:acetic acid:water, 40:10:50) for 10 min followed by Coomassie staining.

RESULTS AND DISCUSSION

In the present investigation, the procedure for two dimensional gel electrophoresis /SDS-PAGE was optimized in order to determine the changes in the various proteins under drought stress in rice shoots. The results showed that under drought stress, expression of various proteins qualitatively and quantitatively changed. Although, without biochemical or sequential identification of various proteins, results obtained in the present study do not allow assuming about the nature of expressed proteins.

Most of the cellular proteins showed similar pattern of alterations in shoot tissues of rice seedlings. Several proteins were below the level of detection in uninduced samples, but showed accumulation with the imposition of drought stress. Expression of six different proteins was up-regulated encountering with 48 h of drought stress in Sahabhagid dhan rice varieties (fig. 1). In control condition also appearance of three proteins was observed but on different positions (fig.2).

Similar results were obtained by **Pandey *et al.*, (2006).** They reported expression of 17 protein spots, out of which 15 protein spots quantitatively down-regulated and 2 spots up-regulated in *Anoectochilus formosanus*, encountering drought stress. **Samlev and Martin (1982)** reported changed electrophoretic spectrum of proteins and decreased quality of chloroplast proteins in the test plants during drought stress.

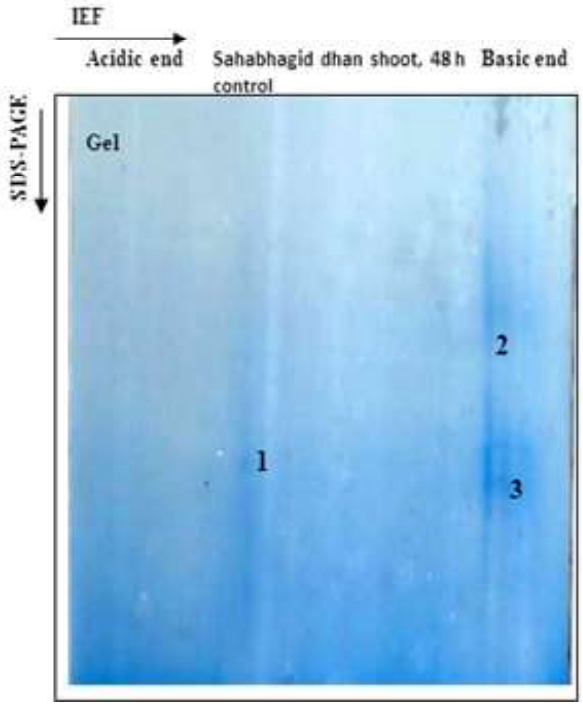


Fig. 1: 2D Analysis of the proteins from shoot tissues of rice (cv Sahabhagid dhan) in uninduced control (48 h). Soluble proteins were first resolved by isoelectric focussing (1.5 mm) gel followed by separation on 12.5% (1.5 mm) uniform concentration SDS-gel. 100mg of total soluble protein fraction was loaded for the analysis. The gels were subsequently stained by using CBB. The duration of treatment (h) are indicated on top of the panel.

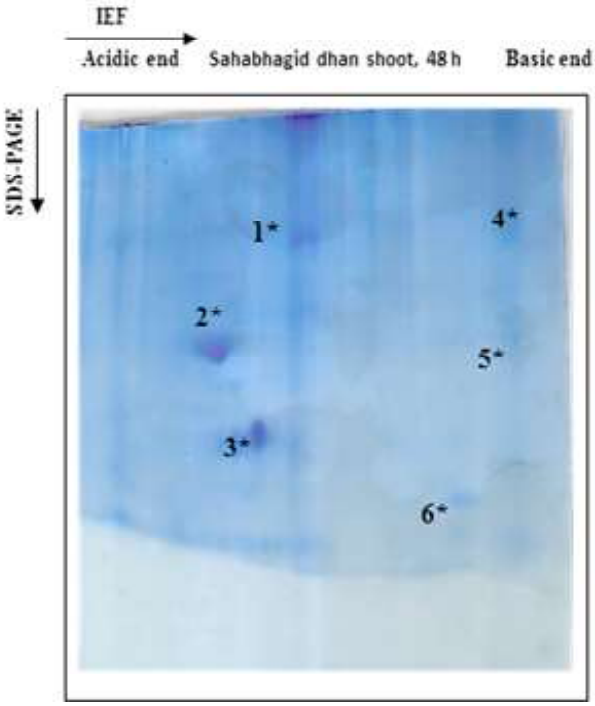


Fig. 2: 2D Analysis of the proteins from shoot tissues of rice (cv Sahabhagid dhan) in response to 48 h of drought stress. Soluble proteins were first resolved by isoelectric focussing (1.5 mm) gel followed by separation on 12.5% (1.5 mm) uniform concentration SDS-gel. 100 g of total soluble protein fraction was loaded for the analysis. The gels were subsequently stained by using CBB. The duration of treatment (h) are indicated on top of the panel.

Abiotic stress negatively influences survival, biomass production and accumulation, and grain yield of most crops. What adaptations (biochemical, physiological, whole-plant level) will allow survival of plants in response to stress regimes? At the heart of all metabolic adaptations are molecular events, and it is the molecular events that we mean when we aim at altering genetics of crops. The future attempts towards improving capacity of drought tolerance through transgenic tools can possibly be further addressed if increased numbers of genes that are involved in drought tolerance are pyramided. There is a need to examine and analyze a much larger number of stress proteins/ genes in order to understand the molecular complexity involved in plant abiotic stress responses.

REFERENCES

1. **Anjum, F., Yaseen, M., Rasul, E., Wahid, A. and Anjum, S.** (2003a). Water stress in barley (*Hordeum vulgare* L.). I. Effect on morphological characters. *Pakistan J. Agric. Sci.*, **40**: 43–44.

2. **Bajaj, S., Targolli, J., Liu, L.F., Ho, THD. and Wu, R.** (1999). Transgenic approaches to increase dehydration-stress tolerance in plants. *Mol Breed* **5:** 493–503.

3. **Bhatt, R.M. and N.K. Srinivasa Rao,** (2005). Influence of pod load response of okra to water stress. *Indian J. Plant Physiol*., **10**: 54– 59.

4. **Bray, E.A.** (1993)**.** Molecular responses to water deficit. *Plant Physiol*. **103**:1035–1040.

5. **Bray, E.A.** (1997). Plant responses to water deficit. *Trends Plant Sci* **2:** 48–54.

6. **Cushman, J.C. and Bohnert, H.J.** (2000). Genomic approaches to plant stress tolerance. *Curr Opin Plant Biol* **3:** 117–124.

7. **Dubey, H., Grover, A., Bhatia, G. and Pasha, S.** (2003). Proteome maps of flood-tolerant FR 13A and flood-sensitive IR 54 rice types depicting proteins associated

with O2-deprivation stress and recovery regimes. *Current Sci,* **84**(1): 83- 89.

8. **Han, B. and A.R. Kermode.** (1996). Dehydrin-like proteins in castor bean seeds and seedlings are differentially produced in responses to ABA and water-deficit-related stresses. *J. Exp. Bot.* **47**:933–939.

9. **Hasegawa, P.M., Bressan, R.A., Zhu, J.K. and Bohnert, H.J.** (2000). Plant cellular and molecular responses to high salinity. *Annu Rev Plant Physiol Plant Mol Biol* **51:** 463–499.

10. **Hirasawa, T.** (1999). Physiological characterization of rice plant for tolerance of water deficit. In: Ito O, O'Toole JC, Hardy B, editors. Genetic improvement of rice for water-limited environments. Los Ba˜nos, Philippines: International Rice Research Institute; p. 89–98.

11. **Kusaka, M., M. Ohta and T. Fujimura,** (2005). Contribution of inorganic components to osmotic adjustment and leaf folding for drought tolerance in pearl millet. *Physiol. Plant.*, **125**: 474–489.

12. **Levitt, J.** (1980). Responses of Plants to Environmental Stress, Ed 2. Academic Press, New York.

13. **Pandey, D.M., Wu, Z.R., Hahn, E.J. and Peak, K.Y.** (2006). Drought effect on electrophoretic protein pattern of *Anoectochilus formosanus*. *Scientia Horti*, **107**: 205-209.

14. **Riccardi, F., Gazeau, P., Vienne, D.V. and Zivy. M.** (1998). Protein changes in responses to progressive water deficit in maize. *Plant Physiol*. **117**:1253–1263.

15. **Samlev, K.S. and Martin, K.G.** (1982). The ultrastructure of chloroplasts of cotton under water deficit. Fiziol. Biokhim. Kul't. Rast. **14**:346-349.

16. **Shao H.B., Chu,L.Y., Shao, M.A., Abdul Jaleel, C. and Hong-Mei, M.** (2008). Higher plant antioxidants and redox signaling under environmental stresses. *Comp. Rend. Biol.*, **331**: 433-441.

17. **Shinozaki, K. and Yamaguchi-Shinozaki, K.** (2000). Molecular responses to dehydration and low temperature: differences and cross-talk between two stress signaling pathways. *Curr Opin Plant Biol* **3:** 217–223.

18. **Tabaeizadeh, Z.** (1998). Drought-induced responses in plant cells. *Int Rev Cytol,* **182**: 193-247.

19. **Thomashow, M.F.** (1999). Plant cold acclimation: freezing tolerance genes and regulatory mechanisms. *Annu Rev Plant Physiol Plant Mol Biol* **50:** 571–599.

20. **Wahid A., Gelani S., Ashraf M. and Foolad, M.R.** (2007). Heat tolerance in plants: an overview, *Environ. Exp. Bot.* **61**: 199–223.

21. **Zhan,g .JX., Klueva, N.Y., Wang, Z., Wu, R., Ho, TH., Nguyen, H.T. and Ho, THD.** (2000). *Genetic* engineering for abiotic stress resistance in crop plants. *In Vitro Cell Dev Biol Plant* **36:** 108–114.

22. **Zivy, M., Thiellement, H., de Vienne, D. and Hofmann, J. P.** (1983). Study on nuclear and cytoplasmic genome expression in wheat by two-dimensional electrophoresis. 1. First results on 18 alloplasmic lines. *Theor Appl Genet*. **66**:1–7.

23. **Zombori, Z., Jancsó, M., Zvara, A., Pauk, J. and Györgyey, J.** (2008). Investigation of the effect of drought stress on the rice Transcriptome. *Acta Biologica Szegediens* **52**(1):143-145.

***